

Econometrics III

Bachelor in Econometrics and Data Science

Bachelor in Econometrics and Operations Research

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Doing Econometrics at Vrije Universiteit Amsterdam

Week 1

Multiple Equation Regression

Welcome to Econometrics III

The Econometrics III team:

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This Course: Econometrics III

1. Multiple equations in regression models (week 1)
2. Panel data models (weeks 2 and 3)
 - ▷ Introduction and Random Effects model
 - ▷ Fixed Effects model and some extensions
3. Vector autoregressive models (weeks 4 and 5)
 - ▷ Stable VAR(1), dynamic properties, forecasting
 - ▷ Granger causality, impulse response, estimation, Stable VAR(p)
4. Non-stationary, cointegration, vector error correction (weeks 5 and 6)
5. Principal components, dynamic factor models (week 6)

Course organization

- ▶ Sundays: slides, exercises, assignments appear on Canvas
- ▶ Mondays: lectures 13:30 in Room HG-10A00
Notice: Week 1: no lecture on Monday (2 February)
- ▶ Tuesdays: lectures 15:30 in Room NU-Theatre 8
- ▶ Wednesdays: tutorial classes, see Canvas
- ▶ Thursdays: assignment meetings, for each group, one student, on-line booking system is active, see Canvas
- ▶ Fridays: self-study

All subject to changes, and for rooms of tutorial and computer classes, please consult <http://rooster.vu.nl> and follow us on Canvas.

Your Econometrics III Weekly Schedule

- ▶ Mondays: lecture and self-study
- ▶ Tuesday: lecture, self-study and exercises
- ▶ Wednesday: tutorial and exercises
- ▶ Thursday: assignment meetings
- ▶ Friday: exercises, assignment and self-study

Econometrics III Assignments

There are three assignments for this course:

- ▶ Assignment 1 is on panel data models, slides `Ect3_panel_2025.pdf`
- ▶ Assignment 2 is on vector autoregressive models, slides `Ect3_VAR_2025.pdf`
- ▶ Assignment 3 is on cointegration and vector error correction models, slides `Ect3_VECM_2025.pdf`

The assignments are done by groups of four students.

The composition of a group must be confirmed via Canvas.

Econometrics III Assignments

- ▶ Each assignment is introduced by a pre-assignment for which you receive the code / output (after a week of its release)
- ▶ The work for the main assignment must be done in a group of four students (± 1 , after approval of Yannick)
- ▶ The same group must work on all three assignments of the course
- ▶ The assignments must be submitted via Canvas before the deadline, see above, no discussion, no excuses
- ▶ You can re-submit the work at Canvas endlessly until the deadline
- ▶ All students in group lose all credits of a non-deliverable assignment
- ▶ Attendance of assignment meeting is for 1 or 2 representatives of group, on-line booking system is in place

Econometrics III Exam Grading policy

► Three assignments:

- ▷ Assignment 1 on panel data : 20 credits
- ▷ Assignment 2 on VAR models : 20 credits
- ▷ Assignment 3 on VECM model : 10 credits

NOTE: a minimum of 20 credits must be obtained.

If less than 20 credits in total of all assignments is obtained, each student of the group receive 0 credits. Maximum is 50 credits.

► Exam: multiple choice exam, 13 questions, each question 4 credits.

NOTE: Minimum of 24 credits must be obtained (6 correct answers). When having less than 24 credits, the score is reduced to 0 credits. You can obtain a maximum of 52 credits.

► Total grade: 100-102 credits is 10 grade, 54 credits or less is FAIL, 55 credits or more is PASS with grade $\max(6, \#credits/10)$.

Any questions ?

Week 1 – Multiple equations in regression

- ▶ Multiple equations in regressions
- ▶ Matrix Algebra
- ▶ Seemingly unrelated regression equations
- ▶ Generalized least squares
- ▶ Feasible generalized least squares
- ▶ Pooled regressions

Regression

Regression model in vector notation:

$$y = X\beta + \varepsilon,$$

with obvious notation.

We have $n \times 1$ vector y , $n \times k$ matrix X and $n \times 1$ vector ε with

- ▶ n is number of realizations from some stochastic process
- ▶ y collects the actual realizations for some variable
- ▶ X contains the corresponding k covariates for each realization
- ▶ ε represent the noise associated with the realizations

The regression model and the variables y , X and ε are subject to the Gauss-Markov assumptions.

For a given regression model $y = X\beta + \varepsilon$, we can estimate β by the ordinary least squares method:

$$\hat{\beta} = (X'X)^{-1}X'y.$$

This regression estimator has optimal properties, certainly when Gauss-Markov assumptions apply, which are discussed in-depth in Econometrics I and II.

The vector y contains realizations for one specific variable:
the dependent variable.

Typically, related variables can also be of interest and can provide better insights in an empirical study.

Why not analyze multiple dependent variables simultaneously ?

It is mostly for the purpose to gain more insights and improve estimation efficiency (more precise estimates).

Food prices and natural disasters in Peru

This topic is relevant in current discussions on the economic implications of climate change.

Relevant for government and inter-governmental agencies, insurance and investment business, international organizations, NGOs, etc.

Collecting good data is of key importance.

But also, to carry out good econometric analyses is of utmost importance !!

Our data set: cross-section / spatial

We have **Food prices**

- ▶ of 14 different products:
banana, carrot, chicken, egg, garlic, lemon, milk, oil, onion, orange, papaya, potato, rice, sugar,
- ▶ which can be grouped as
Vegetables, Fruit, Meat, Processed food
- ▶ or alternatively as
High/Low Calory OR Durable/Perishable
- ▶ from 24 cities:
Ayacucho, Cajamarca, Cusco, Huancavelica, Moquegua, Pasco, etc.
- ▶ which can be grouped in four regions of Peru:
North, South, Center, Jungle.

Our data set: time series

We have **Food prices**, with repeated measurements (realizations) over time

- ▶ for all $14 \times 24 = 336$ products/cities
- ▶ we have monthly observations for 16 consecutive years (2003, 2004, ..., 2018)
- ▶ for each product/city, we have $16 \times 12 = 192$ monthly measurements

Hence, we have a big data set of prices : $336 \times 192 = 64,512$ realizations or data points.

Our data set: covariates

We have **Disaster Costs**, paid by government

- ▶ for 22 different disaster categories:
Volcanic Activity, Avalanche, Cold, Pollution, Landslides, Explosion, Forest Fire, Storm Surge, Drought Earthquake, and more
- ▶ for all 24 cities
- ▶ for all months in 2003, 2004, ..., 2018

We assume that Disaster Costs are exogenous variables.

Research Questions

- ▶ Do occurrences of disasters have an impact on food prices ?
- ▶ Which disasters have a positive impact on food prices and which ones have a negative impact ?
- ▶ Do similar food categories have similar effects or opposite effects due to substitution ?
- ▶ Do disasters impact prices via the supply (producers, transport, groceries) or demand (business, consumer) side of the price equation?
- ▶ Many more

After this course, you will be able to carry out the econometrics that addresses these research questions!!

Single Regression Equation

To investigate whether disasters (total costs) have an impact on the food price of *banana*, in the year of 2018, we do as follows.

We compute the average banana price in 2018 for each city, and place the $n = 24$ averages in the $n \times 1$ data vector y .

The corresponding X matrix has a first column of ones (constant/intercept) and a second column with total disaster costs paid for each city in 2018.

Then, the single regression model

$$y = X\beta + \varepsilon,$$

appears to be valid to address the research question above.

Two Regression Equations

The same research question can be applied to *papaya*.

We repeat the constructions for y and X , but now for papaya.
Then, the single regression model

$$y = X\beta + \varepsilon,$$

addresses again the research question, but now for papaya.

To distinguish the two regression equations, we can write

$$y_1 = X_1\beta_1 + \varepsilon_1, \quad y_2 = X_2\beta_2 + \varepsilon_2,$$

or

$$y_i = X_i\beta_i + \varepsilon_i,$$

for $i = 1$ (*banana*), 2 (*papaya*).

Multiple Regression Equations

More generally, we can consider the i th regression equation

$$y_i = X_i\beta_i + \varepsilon_i,$$

as part of a larger set of “regressions” for all 14 products.

We then have a multiple set of regression equations

$$y_i = X_i\beta_i + \varepsilon_i,$$

for $i = 1, \dots, N$, with $N = 14$ (in our illustration of 14 products).

We can analyze each regression equation separately in this “system” and apply the ordinary least squares method.

However, in our illustration (and in so many others, for that matter) the food prizes of the 14 products may have much in common.

Simultaneous regression analysis

In a multiple set of regression equations

$$y_i = X_i\beta_i + \varepsilon_i,$$

for $i = 1, \dots, N$, where variables can be related to each other, we may wish to analyze the multiple set of regression equations *simultaneously*, as if they are part of a *system of equations*.

We emphasize again that y_i is a vector of observations for the i th variable, for $i = 1, \dots, N$. Further, X_i is a matrix of observations for k explanatory variables (including intercept). The actual values of, say, X_1 and X_2 can be different, but can also be the same.

Example different X_i 's : sales/production of banana and papaya;

Example same X_i 's : total disaster costs in Peru, or total income.

Simultaneous regression analysis

In this multiple set of regression equations

$$y_i = X_i\beta_i + \varepsilon_i,$$

for $i = 1, \dots, N$, the *simultaneous* estimation of the regression coefficients in this *system of equations* can be rather challenging.

However, simultaneous estimation can still rely on least squares methods, but matrix and vector dimensions increase rapidly.

To design effective methods, some more advanced knowledge of *matrix algebra* is useful.

Matrix Algebra

- ▶ Many relevant matrix algebra results can be found online.
- ▶ To obtain a comprehensive introduction: the Luetkepohl book, Appendix A, particularly A.11, A.12, and A.13.
- ▶ However, the main reference on all relevant aspects of matrix algebra in econometrics is book of Abadir and Magnus (2005) with the fitting title “Matrix Algebra”.
- ▶ When your “Matrix Algebra” skills are a bit rusty, please refresh your knowledge and make sure you have reference material at your disposal.
- ▶ Next, we introduce vec notation and Kronecker products.

vec operator

Let $X = [x_1, \dots, x_K]$ be any $n \times K$ matrix, for all $n, K \in \mathbb{N}^+$, with x_i the i th column of X , for $i = 1, \dots, K$.

Then the operator $\text{vec}(X)$ stacks the subsequent columns of X in an $nK \times 1$ vector, from first column to last.

$$\text{vec}(X) = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_K \end{pmatrix}.$$

Example:

$$X = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad \text{vec}(X) = \begin{pmatrix} 1 \\ 4 \\ 2 \\ 5 \\ 3 \\ 6 \end{pmatrix}$$

Kronecker product

Let A be an $m \times n$ matrix and let B be a $p \times q$ matrix.

The $mp \times nq$ matrix

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & a_{ij}B & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix},$$

is defined as the Kronecker product of A and B , for $i = 1, \dots, m$ and $j = 1, \dots, n$.

Kronecker properties

We will make repeated use of Kronecker properties (here appropriate matrix dimensions are assumed):

$$(K1) \quad (A \otimes B)' = A' \otimes B'$$

$$(K2) \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD)$$

$$(K3) \quad (A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$$

$$(K4) \quad \text{vec}(ABC) = (C' \otimes I)\text{vec}(B)$$

$$(K5) \quad \text{vec}(AB) = (I \otimes A)\text{vec}(B) = (B' \otimes I)\text{vec}(A)$$

$$(K6) \quad \text{vec}(B')'\text{vec}(A) = \text{vec}(A')'\text{vec}(B) = \text{tr}(BA) = \text{tr}(AB)$$

$$(K7) \quad \det(A \otimes B) = \det(A)^n \times \det(B)^m \text{ for } m \times m \text{ matrix } A \text{ and } n \times n \text{ matrix } B$$

Most of these properties can be verified or derived easily.

Proof [K4] $\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$

Let matrices A , B and C be such that product ABC exists and assume that matrix C has r rows.

The j th column of ABC corresponds to the j th sub-vector of $\text{vec}(ABC)$, and can be expressed as $ABC e_j$ where e_j is the j th column of the identity matrix, that is e_j is a column vector of zeros except its j th element equals 1. It follows that

$$I_r = \sum_{i=1}^r e_i e_i', \quad \text{for } r \times r \text{ identity matrix } I_r.$$

It is trivial that $ABC = AB I_r C$ and hence $ABC e_j$ is given by

$$AB \left(\sum_{i=1}^r e_i e_i' \right) C e_j = AB \sum_{i=1}^r e_i c_{ij} = \sum_{i=1}^r AB e_i c_{ij} = \sum_{i=1}^r c_{ij} AB e_i,$$

where scalar $c_{ij} = e_i' C e_j$ is the (i, j) element of matrix C .

Proof [K4] $\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$

The j th column of ABC can now be expressed as

$$\sum_{i=1}^r c_{ij} AB e_i = (c_{1j}A, c_{2j}A, \dots, c_{rj}A) \begin{pmatrix} B e_1 \\ B e_2 \\ \vdots \\ B e_r \end{pmatrix}$$

and this is also the expression of the j th sub-vector of $(C' \otimes A)\text{vec}(B)$.

Result applies to any column j of ABC and hence completes the proof.

Abadir and Magnus (2005, p282, Exercise 10.18) provide a slightly different proof.

Can you prove the other Kronecker properties ?

Multiple regression equations

Let us return to the multiple set of regression equations

$$y_i = X_i\beta_i + \varepsilon_i,$$

for $i = 1, \dots, N$, where the variables i can be related to each other.

The vector ε_i contains n noise terms which we assume (for now) to be *normally, independent and identically* distributed (NID).

It implies that

$$\varepsilon_i \sim N(0, \sigma_i^2 I_n),$$

where N is the multivariate normal distribution, I_n is the $n \times n$ identity matrix and σ_i^2 is the variance which can be different for a different equation i .

Hence, the Gauss-Markov assumptions apply to each equation i .

Seemingly Unrelated Regression (SUR)

To facilitate the joint (or simultaneous) analysis of all regression equations

$$y_i = X_i\beta_i + \varepsilon_i,$$

for $i = 1, \dots, N$, we express all equations in one system:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & X_N \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_N \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{pmatrix}$$

or $y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \Sigma)$.

Here, y , X , β , ε are implicitly defined. What can you say about Σ ?

The $Nn \times Nn$ matrix Σ is the variance-covariance matrix for ε .

$$\Sigma = \mathbb{E}(\varepsilon\varepsilon') = \begin{pmatrix} \mathbb{E}(\varepsilon_1\varepsilon_1') & \mathbb{E}(\varepsilon_1\varepsilon_2') & \cdots & \mathbb{E}(\varepsilon_1\varepsilon_N') \\ \mathbb{E}(\varepsilon_2\varepsilon_1') & \mathbb{E}(\varepsilon_2\varepsilon_2') & \cdots & \mathbb{E}(\varepsilon_2\varepsilon_N') \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}(\varepsilon_N\varepsilon_1') & \mathbb{E}(\varepsilon_N\varepsilon_2') & \cdots & \mathbb{E}(\varepsilon_N\varepsilon_N') \end{pmatrix}$$

We have assumed homoskedastic disturbances for each equation, hence $\mathbb{E}(\varepsilon_i\varepsilon_i') = \sigma_i^2 I_n$, for $i = 1, \dots, N$.

Similarly, we can assume across equations that $\mathbb{E}(\varepsilon_i\varepsilon_j') = \sigma_{ij}I_n$, for $i, j = 1, \dots, N$ with $i \neq j$.

We assume *nonzero correlation* between corresponding disturbances (same city) in the i th and j th equations and *zero correlation* between non-corresponding disturbances (different cities).

Illustration for banana and papaya

Consider the two regression equations for banana and papaya, with $n = 24$ observations (from 24 cities):

$$y_1 = X_1\beta_1 + \varepsilon_1, \quad y_2 = X_2\beta_2 + \varepsilon_2,$$

with assumptions $\varepsilon_1 \sim N(0, \sigma_1^2 I_n)$ and $\varepsilon_2 \sim N(0, \sigma_2^2 I_n)$.

A system of (unrelated) regression equations is

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} X_1 & 0 \\ 0 & X_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix}, \quad \mathbb{E}(\varepsilon) = 0, \quad \mathbb{E}(\varepsilon\varepsilon') = \Sigma,$$

with cases

$$\Sigma = \begin{bmatrix} \sigma_1^2 I_n & 0 \\ 0 & \sigma_2^2 I_n \end{bmatrix}, \quad \text{or} \quad \Sigma = \begin{bmatrix} \sigma_1^2 I_n & \sigma_{12} I_n \\ \sigma_{12} I_n & \sigma_2^2 I_n \end{bmatrix}.$$

unrelated regressions

seemingly unrelated regressions

In the SUR $y = X\beta + \varepsilon$, with $\varepsilon \sim N(0, \Sigma)$, we typically have

$$\begin{aligned}\Sigma &= \begin{pmatrix} \sigma_1^2 I_n & \sigma_{12} I_n & \cdots & \sigma_{1N} I_n \\ \sigma_{21} I_n & \sigma_2^2 I_n & \cdots & \sigma_{2N} I_n \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{N1} I_n & \sigma_{N2} I_n & \cdots & \sigma_N^2 I_n \end{pmatrix} \\ &= \Omega \otimes I_n,\end{aligned}$$

$$\Omega = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{N1} & \sigma_{N2} & \cdots & \sigma_N^2 \end{pmatrix},$$

where $\otimes$ is the Kronecker product and $\sigma_{ij} \equiv \sigma_{ji}$, for $i \neq j$, is the noise covariance for the two variable equations i and j .

Seemingly Unrelated Regression (SUR)

The SUR is properly defined as

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & X_N \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_N \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{pmatrix}.$$

Here, the equations have different X s and different β s, and are in this sense unrelated. However, the noise vectors may show dependence across different equations. When we want to allow for this dependence, the *seemingly* enters its meaning in a SUR.

Generalized Least Squares (GLS) Estimation

Given the SUR

$$y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \Sigma),$$

where $\Sigma = \Omega \otimes I_n$ is clearly non-diagonal, we should apply GLS:

$$\hat{\beta}_{\text{GLS}} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}y.$$

Given the potential dimensions of y , X , β and Σ , these necessary computations can be somewhat involved.

The GLS estimator has optimal properties, certainly under Gauss-Markov (or similar / related) assumptions.

Seemingly Unrelated Regressions, same X

We can impose more structure to a SUR equation which can lead to some amazing simplifications for GLS.

Let us consider the same underlying system of equations

$$y_i = X_i \beta_i + \varepsilon_i, \quad i = 1, \dots, N,$$

but now with the same set of explanatory variables, that is $X_i = \bar{X}$, for all equations, for all $i = 1, \dots, N$. We then obtain

$$y_i = \bar{X} \beta_i + \varepsilon_i, \quad i = 1, \dots, N.$$

For our illustration, this structure is very appropriate: each food price (i) is subject to the same disasters and hence the X s are the same; disasters can have different impacts on different food prices and hence the β s are different for different equations.

Seemingly Unrelated Regressions, same X

The SUR with same X is defined as

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} \bar{X} & 0 & \cdots & 0 \\ 0 & \bar{X} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \bar{X} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_N \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{pmatrix},$$

or, similarly,

$$y = [I_N \otimes \bar{X}] \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_N \end{pmatrix} + \varepsilon, \quad \varepsilon \sim N(0, \Sigma),$$

or

$$y = [I_N \otimes \bar{X}] \beta + \varepsilon, \quad \varepsilon \sim N(0, \Sigma), \quad \Sigma = \Omega \otimes I_n.$$

SUR Estimation, same X

Given the model

$$y = [I_N \otimes \bar{X}] \beta + \varepsilon, \quad \varepsilon \sim N(0, \Sigma), \quad \Sigma = \Omega \otimes I_n,$$

the GLS estimate is given by

$$\hat{\beta}_{\text{GLS}} = ([I_N \otimes \bar{X}]' \Sigma^{-1} [I_N \otimes \bar{X}])^{-1} [I_N \otimes \bar{X}]' \Sigma^{-1} y,$$

and by applying some Kronecker properties, we obtain

$$\begin{aligned} [I_N \otimes \bar{X}]' \Sigma^{-1} [I_N \otimes \bar{X}] &= [I_N \otimes \bar{X}]' [\Omega \otimes I_n]^{-1} [I_N \otimes \bar{X}] \\ &= [I_N \otimes \bar{X}'] [\Omega^{-1} \otimes I_n] [I_N \otimes \bar{X}] \\ &= \Omega^{-1} \otimes \bar{X}' \bar{X}, \end{aligned}$$

such that

$$\hat{\beta}_{\text{GLS}} = \left(\Omega^{-1} \otimes \bar{X}' \bar{X} \right)^{-1} [\Omega^{-1} \otimes \bar{X}'] y.$$

SUR Estimation, same X

Given the GLS expression

$$\hat{\beta}_{\text{GLS}} = \left(\Omega^{-1} \otimes \overline{X}'\overline{X} \right)^{-1} [\Omega^{-1} \otimes \overline{X}']y.$$

and after applying some further Kronecker properties, we obtain

$$\hat{\beta}_{\text{GLS}} = \left(\Omega \otimes (\overline{X}'\overline{X})^{-1} \right) [\Omega^{-1} \otimes \overline{X}']y = (I_N \otimes (\overline{X}'\overline{X})^{-1}\overline{X}')y.$$

This is a very elegant result.

It implies that for a SUR with same X s, you can obtain the GLS estimate by simply applying ordinary least squares to each equation. No joint estimation is needed !

SUR system with different X 's

Returning to the original SUR

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} X_1 & 0 & \cdots & 0 \\ 0 & X_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & X_N \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_N \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{pmatrix}.$$

In many cases, the assumptions that $X_i = \bar{X}$, for $i = 1, \dots, N$, are somewhat or even quite restrictive !!

Equation-by-equation estimation can still be useful and is used in the context of *feasible* GLS.

Feasible GLS Estimation

For a SUR with $Nn \times 1$ vector y , $Nn \times Nk$ matrix X and $Nk \times 1$ vector β ,

$$y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \Sigma),$$

with $Nn \times Nn$ variance matrix $\Sigma = \Omega \otimes I_n$, we can apply GLS:

$$\hat{\beta}_{\text{GLS}} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}y.$$

The computation of $\hat{\beta}_{\text{GLS}}$ requires a known variance matrix Ω . This is not realistic in the real world. A practical solution is the two-step procedure of feasible GLS:

1. Estimate equation-by-equation, save the N set of least squares residuals and construct estimate $\hat{\Omega}$;
2. Replace Ω by $\hat{\Omega}$ and compute $\hat{\beta}_{\text{GLS}}$ accordingly, that is $\hat{\beta}_{\text{FGLS}}$.

Least squares

Carry out ordinary least squares, equation-by-equation, possibly with different X s.

Given the least squares estimates $\hat{\beta}_{i,LS}$ for $i = 1, \dots, N$, we compute the least squares residuals as

$$e_i = y_i - X_i \hat{\beta}_{i,LS}, \quad i = 1, \dots, N.$$

Estimates of the elements of the variance-covariance matrix Ω are given by

$$\hat{\sigma}_i^2 = \frac{e_i' e_i}{n - K}, \quad \hat{\sigma}_{ij} = \frac{e_i' e_j}{n - K},$$

for $i, j = 1, \dots, N$, with $i \neq j$, which provide the estimate $\hat{\Omega}$.

Feasible GLS

Given the estimate $\hat{\Omega}$, we can carry out feasible GLS (FGLS), for general case with possibly different X s, by computing

$$\hat{\beta}_{\text{FGLS}} = \left(X' \left[\hat{\Omega} \otimes I_N \right]^{-1} X \right)^{-1} X' \left[\hat{\Omega} \otimes I_N \right]^{-1} y.$$

with the corresponding “feasible” variance matrix of $\hat{\beta}_{\text{FGLS}}$ given by

$$\hat{V} = \left[\frac{1}{n} X' \left(\hat{\Omega}^{-1} \otimes I_N \right) X \right]^{-1}.$$

This expression is different when compared to case of GLS where Ω is assumed to be known, that is $\hat{V} = \Omega \otimes (\bar{X}'\bar{X})^{-1}$.

Feasible GLS

FGLS is effectively a *plug-in* estimate. However, it can be shown that

$$\hat{\beta}_{\text{FGLS}} = \left(X' \left[\hat{\Omega} \otimes I_N \right]^{-1} X \right)^{-1} X' \left[\hat{\Omega} \otimes I_N \right]^{-1} y.$$

is consistent, as long as $\hat{\Omega}$ is a consistent estimate of Ω ,

$$\text{plim } \hat{\beta}_{\text{FGLS}} = \beta,$$

and asymptotic normally distributed,

$$\sqrt{n} \left(\hat{\beta}_{\text{FGLS}} - \beta \right) \xrightarrow{d} N(0, V),$$

where

$$V = \text{plim} \left(\frac{1}{n} X' \left[\hat{\Omega}^{-1} \otimes I_N \right] X \right)^{-1} = \text{plim } \hat{V}.$$

Feasible GLS

It implies that the FGLS estimate

$$\hat{\beta}_{\text{FGLS}} = \left(X' \left[\hat{\Omega} \otimes I_N \right]^{-1} X \right)^{-1} X' \left[\hat{\Omega} \otimes I_N \right]^{-1} y.$$

can be used for inference on the parameter vector β based on the approximate normality of $\hat{\beta}_{\text{FGLS}}$, that is

$$\hat{\beta}_{\text{FGLS}} \overset{a}{\sim} N\left(\beta, \hat{V}\right).$$

These results have much practical relevance as we can do approximate inference for many models of interest in applied econometrics including panel data models and vector autoregressive models which we will consider in this course.

Pooling the parameter vectors

We have considered the SUR model

$$y_i = X_i \beta_i + \varepsilon_i, \quad i = 1, \dots, N,$$

which we can write in matrix form as

$$y = X\beta + \varepsilon,$$

where $X = \text{diag}(X_1, \dots, X_N)$ and $\beta = (\beta'_1, \dots, \beta'_N)'$.

We have considered the case where all X_i 's are the same for all i 's, that is $X_i \equiv \bar{X}$.

We can further consider “pooled estimation” where we have all β_i 's to be the same, that is $\beta_i = \beta^*$, for all i 's.

Pooling the parameter vectors

Pooling the parameter vectors $\beta_i = \beta^*$, for all i 's, lead to the model

$$y_i = X_i \beta^* + \varepsilon_i, \quad i = 1, \dots, N,$$

or in matrix form as

$$y = X^* \beta^* + \varepsilon, \quad X^* = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix},$$

and possibly we have same X 's with $X_i = \bar{X}$.

So we do not “diag” X s but we are “stacking” X s.

Application of least squares or GLS is straightforward for this model.

Pooled regression forms the basis for panel data models, next !!

What did we do?

- ▶ Setting the stage for regression
- ▶ Introducing multiple set of regression equations
- ▶ Reviewing matrix algebra
- ▶ Introducing SUR
- ▶ SUR with more structure, same X
- ▶ Feasible GLS
- ▶ Pooled estimation

Work to do in this first week

- ▶ Revisit the Econometrics I course material
- ▶ Know matrix algebra: Abadir and Magnus book on your table
- ▶ Reproduce all derivations of these slides

Good luck !!